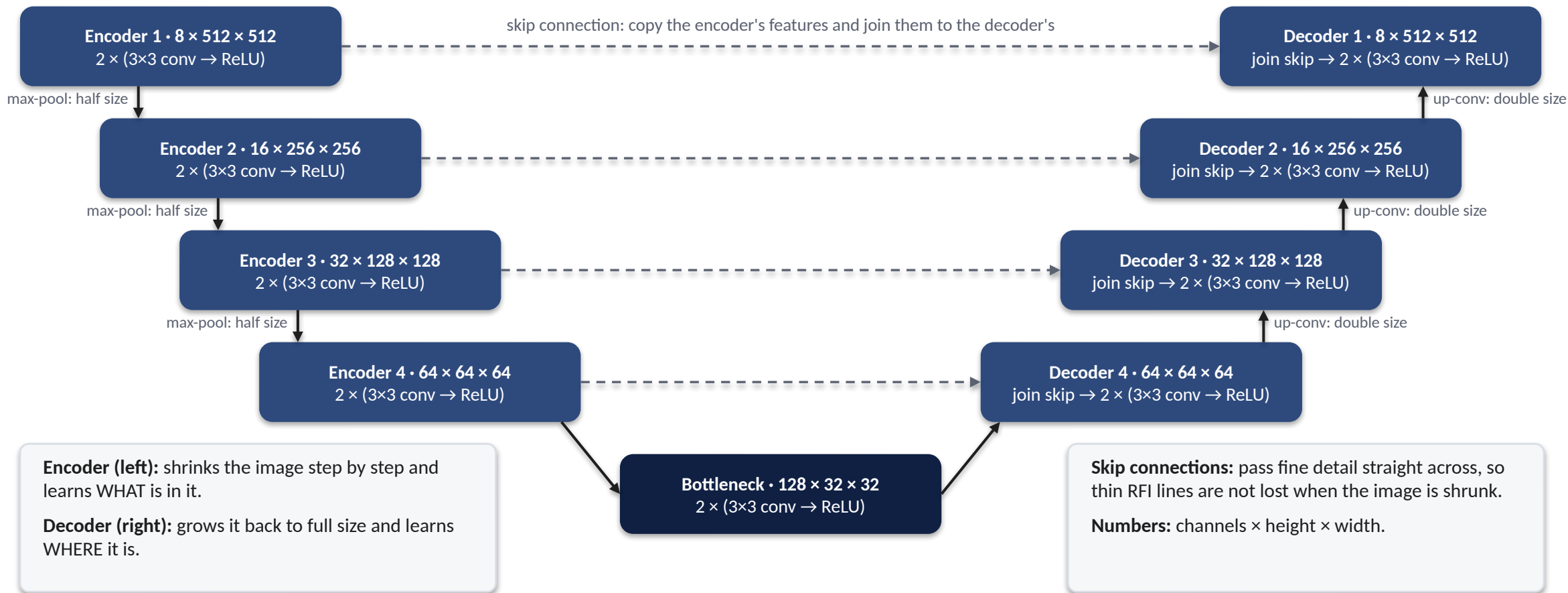


➤ Starting point: a basic U-Net

▼ Input: spectrogram, $1 \times 512 \times 512$ (time $\times$ frequency)

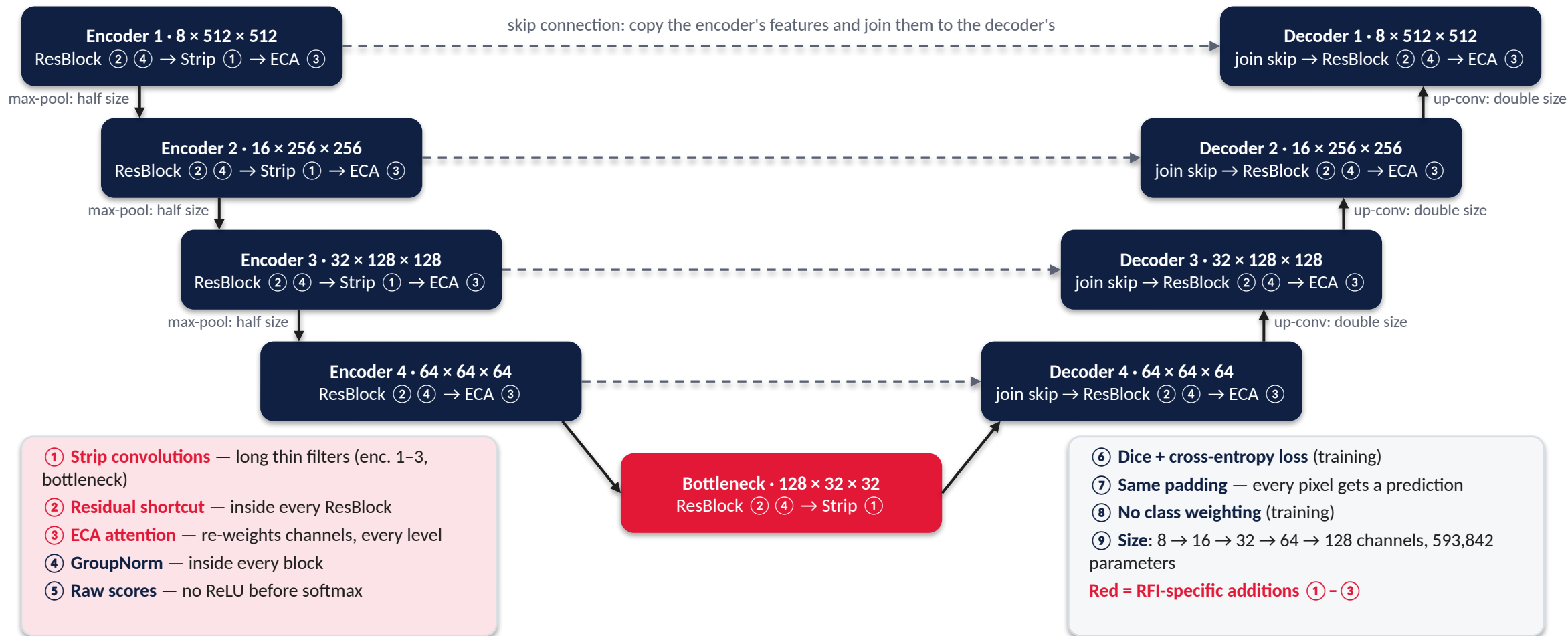
Output: an RFI probability for every pixel ▲



➤ Our model: the same U, with 9 changes

▼ Input: spectrogram, $1 \times 512 \times 512$ — sizes never shrink ⑦

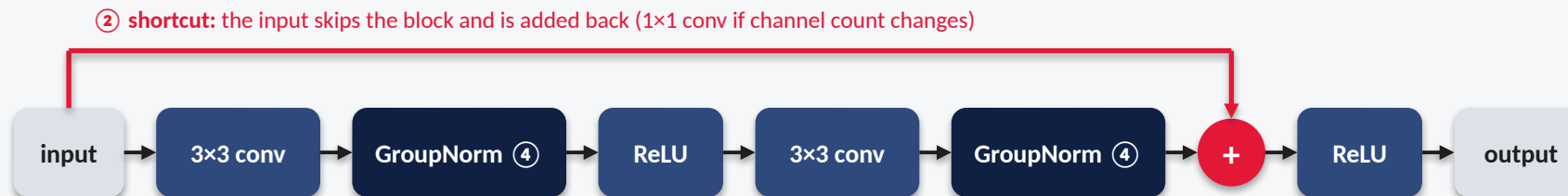
1×1 conv $\rightarrow$ 2 raw scores $\rightarrow$ softmax ⑤ $\rightarrow$ RFI probability ▲



➤ Inside one level, step by step

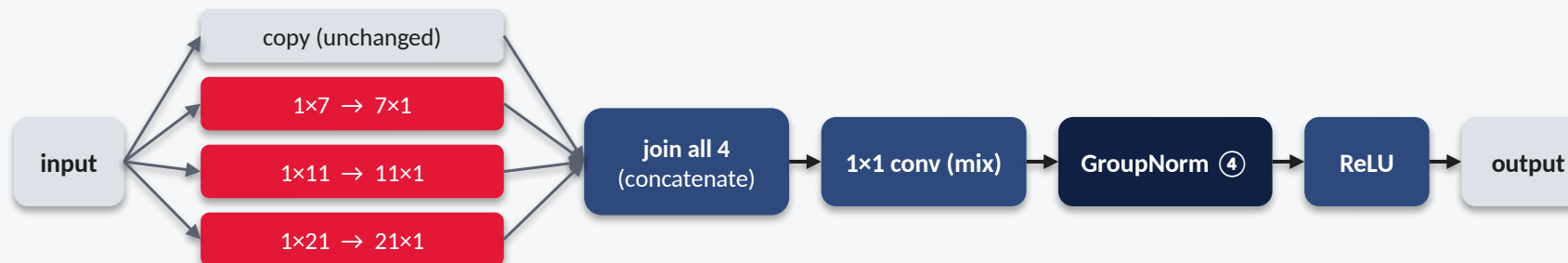
ResBlock ② ④

Two 3×3 convolutions, each tidied by GroupNorm; the input is added back at the end.



Strip module ①

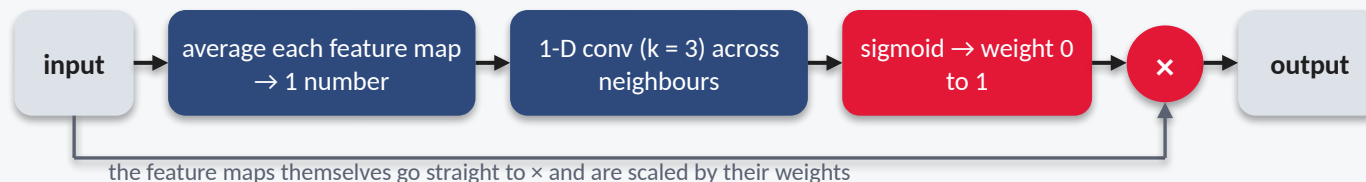
Looks along long thin lines of 7, 11 and 21 pixels, in time and in frequency.



On LOFAR (rows = time): 1×K runs along frequency (catches broadband bursts) · K×1 runs along time (catches narrowband lines)

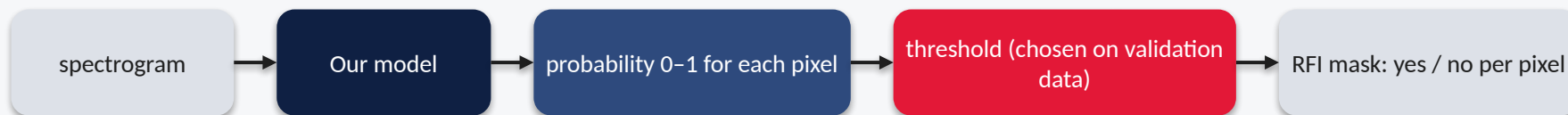
ECA attention ③

Gives each feature map a weight between 0 and 1, then scales it.

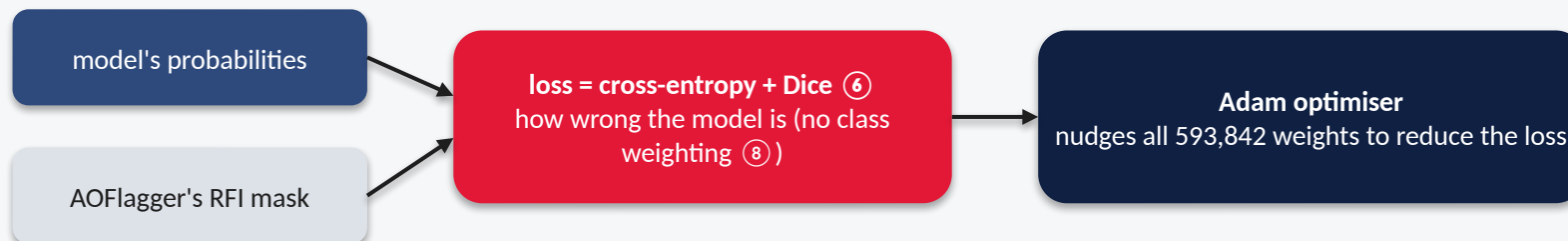


➤ From output to RFI mask, and how it learns

Using the model
after training



Training
repeated 28,000 times



Which parts actually matter? Removing ① ② ③ (strips, shortcuts, attention) changes F1 from 0.660 to 0.658 — not significant (3 runs each, $p = 0.64$).

So the model's strength comes from ④ - ⑨: normalisation, raw output scores, the Dice loss, same padding, no class weighting, and its size and training setup.